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ORIGINAL ARTICLE

Effects of sulphur type and concentration on growth performance, body composition and resistance of juvenile sea cucumber (*Apostichopus japonicus*) subjected to air and low salinity exposure stressors

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Abstract

Supplementation effect of sulphur type and concentration on survival, growth and stress resistance of sea cucumber has not been known. Two-way {2 type sulphur [liquid type (LTS) and nano-powder type (NTS)] ×5 sulphur concentrations (25, 50, 100, 250 and 500 ppm)} ANOVA design was applied. LTS and NTS were included in the control (Con) diet at five sulphur concentrations instead of equal amount of water and cellulose, referred to as the LTS25, LTS50, LTS100, LTS250 and LTS500, and NTS25, NTS50, NTS100, NTS250 and NTS500 diets, respectively. Sea cucumbers were fed with triplicate groups of each diet for 8 weeks. After 8-week feeding trial, sea cucumbers were exposed to 30-h air and 12-h low salinity exposure stressors and cumulative mortality was monitored. Survival was not affected by either sulphur type or sulphur concentration. Weight gain of sea cucumber was affected by both sulphur type and concentration. Sea cucumber fed all NTS diets except for the NTS25 diet achieved better weight gain than that fed the Con and all LTS diets. Chemical composition of sea cucumber was not affected by either dietary sulphur type or sulphur concentration. The cumulative mortality of sea cucumber was affected by both sulphur type and concentration after the 30-h air and 12-h low salinity exposure stressors. In conclusion, NTS was more effective than LTS to improve growth and lower mortality of sea cucumber subjected to the 30-h air and 12-h low salinity exposures. The greatest weight gain and resistance were obtained in sea cucumber fed the NTS500 diet.

KEYWORDS
cumulative mortality, Sea cucumber (*Apostichopus japonicus*), stress endurance, sulphur concentration, sulphur type

1 | INTRODUCTION

Sea cucumber (*Apostichopus japonicus*) is a high valued marine epibenthic echinoderm, having great social and economic importance to the coastal community of different countries including Korea, China and Japan (Oh et al., 2014; Okorie et al., 2008; Ru et al., 2019). The annual aquaculture production of sea cucumber reached to 93 metric tons in Korea in 2019 (KOSIS 2020). Since it is good for human health, the international market demand of sea cucumber for human consumption is in increasing trend (Seo et al., 2011). In

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addition, it has curative properties and been used as traditional medicines in China (Chen, 2004; Jiang et al., 2015). However, over-exploitation causes a serious problem of declining wild production of sea cucumber (Conand, 2004; Uthicke, 2004). The quality (protein and lipid contents) of artificial diet is one of principal factors affecting growth and development of sea cucumber (Seo & Lee, 2011). Nevertheless, the high cost and unavailability of artificial feed are also a key challenge for aquaculture development (Conand, 2004). Therefore, continuous research is needed to develop an artificial diet to increase the sustainable production of sea cucumber.

Several studies have been performed on development of formulated diets for sea cucumber, such as optimum dietary protein and lipid levels (Liao et al., 2017; Seo & Lee, 2011) and carbohydrate level (Xia et al., 2015), dietary inclusion effect of sea mud (Kim et al., 2016), and dietary substitution effect of *Sargassum thunbergii* with different plant ingredients (*Saccharina japonica*, *Undaria pinnatifida*, *Brassica oleracea* L. var. *capitata*, distilled dried grain, fermented soybean and rice straw powder) (Seo et al., 2011) and macroalgae (*S. polycystum* and *Laminaria japonica*) (Liu et al., 2010), but knowledge on artificially nutrition-balanced diet that is capable of improving production of commercially important sea cucumber is still limited (Yuan et al., 2006; Zhou et al., 2006). Since sea cucumber culture in Korea heavily depends on the imported feed from China (Kim et al., 2016), it is highly required to develop practically suitable diet for domestic sea cucumber culture.

As sulphur is an essential component in amino acids, proteins, enzymes, vitamins and other biomolecules, it is considered as an indispensable nutrient for all living kingdom (Kormarnisky et al. 2003). It is also a key component of sulphur-containing amino acids (cysteine, methionine, homocysteine and taurine) (Brosnan & Brosnan, 2006), iron-containing flavoenzymes and other compounds having antioxidant activity, such as lipoic acid, mercaptopropionylglycine, N-acetylcysteine, diallylsulphide, diallyldisulphide and diallyltrisulphide (Atmaca, 2004). Richter et al. (2012) suggested that dietary sulphur might influence lipid metabolism of beef steer.

Methionine is one of the most important sulphur-containing essential amino acids that play an important role in the antioxidative defence and influence physiological processes including growth and immunity and carbohydrate and lipid metabolism of animal (Li et al., 2007, 2009; Metayer et al., 2008). Insufficient methionine (1.6 g methionine/16 g nitrogen) content in diet for juvenile Atlantic salmon (*Salmo salar*) reduced growth and protein gain compared with those of fish fed the diet containing DL-methionine (2.1 g methionine/16 g nitrogen) in the 8-week feeding trial (Espe et al., 2014). Gunathilaka et al. (2019) described that dietary supplementation of taurine increased feed utilization and overall growth performance of red seabream (*Pagrus major*) when fish were fed with one of diets substituting fish meal with soybean meal without taurine and with 1% taurine during in water temperature season. Wilson and Halver (1986) reported that irrespective of a specific amino acid, fish have a requirement for total sulphur amino acid. Sufficient amount of cysteine in diet spares methionine requirement for fish (Farhat, 2014; Goff & Gatlin, 2004) and furthermore methionine and cysteine inclusion may spare dietary taurine requirement as they are the precursors of taurine (Stipanuk & Dominy, 2006; Wang, He, et al., 2014a).

The previous studies reported that supplementation of sulphur-containing antioxidant is beneficial for controlling some human diseases (Ingenbleek & Kimura, 2013; Kim & Seo, 1996; Parcell & Candidate, 2002). Shin et al. (2013) stated that supplementation of 0.2% processed sulphur in chicken feed could improve quality of poultry products and antioxidant activity. Organosulphur compound sulphonyl methane improved growth performance of pigs (Cho et al., 2005). Although studies on the positive effect of administration sulphur on terrestrial animals are available, there is still lack of research in the field of aquatic animals.

The sustainability of aquatic industry mainly depends on the efficient use of available resources to meet the demand for essential nutrition, increase growth performance and reduce the risk of mortality of farmed fish and other aquatic animals caused by biotic and abiotic factors. Sea cucumber is also prone to different abiotic factors affecting its growth, immunity and survival (Hou et al., 2019; Vidolin et al., 2002). Water exchange, evaporation and precipitation are critical factors fluctuating salinity (Wang, Yu, Qin, Dong & Dong, 2014b), whereas transportation, size grading and other farm activities are responsible for air exposure stressor in juvenile sea cucumber culturing (Hou et al., 2019). Sea cucumber subjected to air and low salinity exposures are ones of unavoidable stressors during sea cucumber farming in pond culture. As sea cucumber (*A. japonicus*) tolerates salinity fluctuations when inhabiting the intertidal zone and culturing ponds, it can survive within 22–38 psu, but 30–32 psu for the optimum salinity, respectively (Bai et al., 2015). Scientists reported that improvement in nutrition status of diet could lead to better resistance capacity of aquatic animals against different abiotic stressors (Ansary, Baek, et al., 2019; Ansary, Jeong, et al., 2019; Lee et al., 2016; Liu et al., 2015). Therefore, it is necessary to develop nutritionally rich diet to improve health status of sea cucumber against common abiotic stressors, salinity and air exposure stressors.

The present study was, therefore, conducted to investigate the effects of sulphur type and content in diets on growth, body composition and tolerance ability of sea cucumber against air and low salinity exposures, which are the unavoidable stressors in sea cucumber pond culture affecting survival (production) of the animal.

2 | MATERIALS AND METHODS

2.1 | Experimental animal and rearing condition

Sea cucumber juveniles were purchased from a private hatchery (Namhae-gun, Gyeongsangnam-do, Republic of Korea), transferred and acclimatized to the experimental condition for 2 weeks prior to an initiation the feeding trial. During the acclimatization period, juveniles were fed with commercial feed (the mixture of *Sargassum* sp. and mud).

A total of 1800 juvenile (initial weight of 1.46 ± 0.001 g: mean ±SD) sea cucumber were randomly distributed into indoor 36, 300 L round-shaped fibre-reinforced plastic (FRP) tanks [80 cm (diameter) ×100 cm

(depth)] (50 juvenile sea cucumber/tank) with flow-through system. Each tank has 3 black shelters (25 cm ×20 cm ×2 cm), and the sand-filtered seawater was supplied. Water temperature ranged from 8.7 to 15.5°C (mean ±SD; 12.0 ± 0.16°C) and flow rate was 3 L min^{-1} $tank^{-1}$. A photoperiod of 10 h:14 h (light:dark) cycle was used. The diets were mixed well with seawater in a 50 mL tube and fed to sea cucumber once a day (14:00) at the ratio of 2–4% of biomass for 8 weeks. Aeration and water supply stopped for 3 h after feed supply. Aeration was supplied at 17:00 again, but sand-filtered seawater was again supplied in the morning (09:00) in the following day until the next designated feed supply. To maintain proper water quality for sea cucumber culture, uneaten feed and faeces of sea cucumber in tanks were siphon-cleaned daily. Dead sea cucumbers were removed at daily basis.

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to. No ethical approval was required for this study.

2.2 | Diets formulation and experimental design

Twelve experimental diets were prepared (Table 1). The control (Con) diet contained 40% defatted soybean meal and 10% fish meal as the protein source. Thirty-five per cent macroalgae (*Undaria pinnatifida*:*Saccharina japonica* at the ratio of 1:1) powder was included in the Con diet.

Sulphur types were classified into 2 types: liquid type sulphur (LTS; 22.24% sulphur content) and nano-powder type sulphur (NTS; 99.00% sulphur content and 100 nm particle) supplied from EF Biotech (Bucheon City, Gyeonggi-do, Republic of Korea). Both LTS and NTS are in the forms of processed organic sulphur compounds. The experiment was designed with two-way [2 type sulphur (LTS and NTS) ×5 sulphur concentration (25, 50, 100, 250 and 500 ppm)] ANOVA with triplication. Five different concentrations (25, 50, 100, 250 and 500 ppm) of LTS were included into the Con diet instead of an equal amount of water, referred to as the LTS25, LTS50, LTS100, LTS250 and LTS500 diets respectively. LTS was diluted with distilled water and prepared at 100 ppm LTS stock solution. LTS stock solution was stored in a 500 ml dark bottle at 4°C refrigerator and newly prepared once a week. LTS stock solution was pipetted as the designated concentrations and dispersed into water surface of 300 L FRP tanks.

Five different concentrations (25, 50, 100, 250 and 500 ppm) of NTS were included into the Con diet at the expense of

TABLE 1 Formulation and chemical composition of the experimental diets (%, dry matter basis).

	Experimental diets						
	Con	NTS25	NTS50	NTS100	NTS250	NTS500	PTS100
Ingredient (%)							
Defatted soybean meal	40	40	40	40	40	40	40
Brown fish meal	10	10	10	10	10	10	10
Macroalgae powder[a]	35	35	35	35	35	35	35
Wheat flour	10	10	10	10	10	10	10
Cellulose	1.0000	0.9975	0.9950	0.9900	0.9750	0.9500	0.9900
Nano-powder type sulphur (NTS)[b]	0.0000	0.0025	0.0050	0.0100	0.0250	0.0500	0.0000
Power type sulphur (PTS)	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Vitamin premix[c]	2	2	2	2	2	2	2
Mineral premix[d]	2	2	2	2	2	2	2
Nutrients (%, DM)							
Dry matter	91.8	91.1	91.7	91.2	91.2	91.5	91.3
Crude protein	28.5	28.7	28.7	28.7	28.7	28.9	29.0
Crude lipid	2.9	2.9	2.9	2.9	3.0	3.0	3.0
Ash	19.2	19.2	19.0	19.1	19.3	19.2	19.1
Sulphur	1.70	1.76	1.87	2.23	2.73	2.99	2.37

[a]Macroalgae powder is composed of *Undaria pinnatifida*:*Saccharina japonica* at the ratio of 1:1.

[b]Sulphur was supplied from EF Biotech (Bucheon City, Gyeonggi-do, Republic of Korea), and the same concentration of liquid type sulphur (LTS) was included in the control (Con) diet instead of equal amount of water, referred to as the LTS25, LTS50, LTS100, LTS250 and LTS500 diets respectively.

[c]Vitamin premix contained the following amount which were diluted in cellulose (g/kg mix): L-ascorbic acid, 200; α-tocopheryl acetate, 20; thiamin hydrochloride, 5; riboflavin, 8; pyridoxine, 2; niacin, 40; Ca-D-pantothenate, 12; myo-inositol, 200; D-biotin, 0.4; folic acid, 1.5; p-amino benzoic acid, 20; K3, 4; A, 1.5; D3, 0.003; cyanocobalamin, 0.003.

[d]Mineral premix contained the following ingredients (g/kg mix): NaCl, 7, $MgSO_4.7H_2O$, 105; $NaH_2PO_4.2H_2O$, 175; KH_2PO_4, 224; $CaH_4(PO_4)_2.H_2O$, 140; Ferric citrate, 17.5; $ZnSO_4..7H_2O$, 2.8; Ca-lactate, 21.8; CuCl, 0.2; $AlCl_3.6H_2O$, 0.11; KIO_3, 0.05; $Na_2Se_2O_3$, 0.007; $MnSO_4.H_2O$, 1.4; $CoCl_2.6H_2O$, 0.07.

TABLE 2 Growth performance of juvenile sea cucumber fed the experimental diets for 8 weeks.

Experimental diets	Initial weight (g/sea cucumber)	Final weight (g/sea cucumber)	Survival (%)	Weight gain (g/sea cucumber)	SGR† (%/day)
Con	1.46 ± 0.00	3.68 ± 0.029[e]	99.3 ± 0.67	2.21 ± 0.030[f]	1.64 ± 0.015[f]
LTS25	1.46 ± 0.00	3.67 ± 0.022[e]	100.0 ± 0.00	2.21 ± 0.023[f]	1.65 ± 0.012[f]
LTS50	1.46 ± 0.00	3.68 ± 0.028[e]	100.0 ± 0.00	2.22 ± 0.027[f]	1.65 ± 0.012[f]
LTS100	1.46 ± 0.01	3.71 ± 0.014[e]	100.0 ± 0.00	2.25 ± 0.019[f]	1.66 ± 0.013[f]
LTS250	1.46 ± 0.00	3.71 ± 0.026[e]	100.0 ± 0.00	2.25 ± 0.027[f]	1.66 ± 0.014[f]
LTS500	1.46 ± 0.00	3.73 ± 0.020[e]	100.0 ± 0.00	2.27 ± 0.021[ef]	1.67 ± 0.011[ef]
NTS25	1.46 ± 0.00	3.79 ± 0.021[d]	100.0 ± 0.00	2.33 ± 0.021[de]	1.70 ± 0.009[de]
NTS50	1.46 ± 0.00	3.83 ± 0.025[d]	100.0 ± 0.00	2.37 ± 0.027[d]	1.72 ± 0.015[d]
NTS100	1.46 ± 0.00	3.93 ± 0.015[bc]	100.0 ± 0.00	2.46 ± 0.011[bc]	1.76 ± 0.002[bc]
NTS250	1.46 ± 0.00	3.99 ± 0.022[ab]	100.0 ± 0.00	2.53 ± 0.021[ab]	1.79 ± 0.009[ab]
NTS500	1.46 ± 0.00	4.02 ± 0.011[a]	100.0 ± 0.00	2.56 ± 0.012[a]	1.81 ± 0.005[a]
PTS100	1.46 ± 0.00	3.90 ± 0.014[c]	100.0 ± 0.00	2.44 ± 0.015[c]	1.75 ± 0.007[c]
Two-way ANOVA					
Type	$p > 0.7$	$p < 0.0001$	$p > 0.3$	$p < 0.0001$	$p < 0.0001$
Concentration	$p > 0.6$	$p < 0.0001$	$p > 0.4$	$p < 0.0001$	$p < 0.0001$
Interaction	$p > 0.5$	$p < 0.001$	$p > 0.4$	$p < 0.001$	$p < 0.004$

Values (means of triplicate ±SE) in the same column sharing the same superscript letters are not significantly different ($p > 0.05$).
Specific growth rate (SGR, %/day) = (ln final weight of sea cucumber −ln initial weight of sea cucumber) ×100/days of feeding trial.

cellulose, referred to as the NTS25, NTS50, NTS100, NTS250 and NTS500 diets respectively. To determine the effect of particle size, 100 ppm powder type sulphur (PTS; 99.14% sulphur content and 30–40 μm particle) was included in the Con diet instead of an equal amount of cellulose, referred to as the PTS100 diet. All feed ingredients were sieved with 250 μm mesh sieve (Samwoo Industry Co., Daegu, Republic of Korea) and mixed through well. The experimental NTS and PTS diets were stored in a freezer at −15°C and used in small amount whenever necessary. Each experimental diet was separately held into 50 mL tubes, mixed with seawater well, and then dispersed into the designated 300 L FRP tanks. Sea cucumbers were daily fed with triplicate groups of each diet for 8 weeks.

2.3 | Measurement of growth performance and chemical composition of sea cucumber

After the 8-week feeding trial, sea cucumbers from each tank were collectively weighed to measure weight gain. Ten sea cucumbers from each tank, except for one tank in the Con diet (9 sea cucumber) randomly chosen and sacrificed for the chemical analysis of the whole body. Specific growth rate (SGR) (%/day) was calculated as follows: SGR= (ln final weight of sea cucumber −ln initial weight of sea cucumber ×100/days of feeding trial). The proximate composition of sea cucumber was determined using the standard AOAC (1990) methods. Crude protein (N × 6.25) was analysed by the Kjeldahl method (Buchi, B-324/435/412, Switzerland), an ether-extraction method was used to measure crude lipid content, moisture was measured by oven drying at 105°C for 24 h and ash content was measured by a muffle furnace at 550°C for 4 h.

2.4 | Assessment of tolerance resistance of sea cucumber subjected to air exposure stressor

To evaluate tolerance resistance of sea cucumber to air exposure stressor, 20 sea cucumbers were randomly selected from each tank after the 8-week feeding trial and the rest sea cucumber was transferred to 1, 3-ton rectangular tank to evaluate low salinity stressor test. The seawater was completely drained from the tanks and exposed to air exposure stress for 30 h. Then, the tanks were refilled with sand-filtered seawater and maintained the same conditions as the feeding trial without feed supply. The cumulative mortality was monitored for the following 4 days after the 30-h air exposure stressor. Dead sea cucumber were removed every 3 h for the first 2 day and every 6 h for the remaining 2 day during the 4-day post observation period. In addition, sea cucumber fed the commercial feed in extra tanks for 8 weeks were also prepared to compare dietary inclusion effect of sulphur type and concentrate on mortality.

2.5 | Assessment of tolerance resistance of sea cucumber subjected to low salinity stressor

Twenty sea cucumbers were randomly chosen from each tank at the end of the 8-week feeding trial. A total of 39 round-shaped plastic bottom-screened containers (30 cm in diameter and 8.5 cm in

height) were placed in 1, 3-ton rectangular tank adjusted at 10 psu salinity by mixing tap water with sand-filtered seawater (32 psu). Salinity was measured by using YSI 6-Series Multi Parameter (YSI, Yellow Springs, OH, USA). Sea cucumbers were exposed to low saline water (10 psu) for 24 h, and low saline water was completely drained out. Then, a rectangular tank was refilled with sand-filtered seawater to monitor the effect of low salinity stressor on mortality of sea cucumber for the next 96-h post observation. Sea cucumber was starved during the 96-h post observation after the 24-h low saline exposure stressor. Dead sea cucumbers were removed every 3 h for the first 48 h and 6 h for the remaining 48 h. In addition, sea cucumber fed the commercial feed in extra tanks for 8 weeks were prepared to compare dietary inclusion effect of sulphur type and concentrate on mortality.

2.6 | Statistical analysis

Two-way ANOVA and Duncan's multiple range test (Duncan, 1955) using SPSS version 24.0 (SPSS Inc.) were used to compare the means of treatment. Percentage data were subjected to arcsine-transformation prior to analysis. Additionally, regression analysis of weight gain of sea cucumber and sulphur contents in the NTS diets at the end of the 8-week feeding trial was performed.

3 | RESULTS

3.1 | Growth performance of sea cucumber

Survival of sea cucumber ranged from 99.3 to 100.0% and was not significantly affected by either sulphur type ($p > 0.3$) or sulphur concentration ($p > 0.4$) (Table 2). However, weight gain and SGR of sea cucumber were significantly affected by both sulphur type ($p < 0.0001$) and concentration ($p < 0.0001$). Their significant ($p < 0.001$ and $p < 0.004$, respectively) interaction was also observed. Regardless of sulphur concentration, except for the LTS500 diet, weight gain and SGR of sea cucumber fed all NTS diets and the PTS100 diet were significantly ($p < 0.05$) greater than those of sea cucumber fed all LTS diets and the Con diet. The NTS diets achieved greater weight gain and SGR of sea cucumber than those of sea cucumber fed the LTS diet at the same concentration of sulphur content. The following correlationship between weight gain of sea cucumber (Y) and concentration of sulphur in the NTS diets (X) were observed: Y (Weight gain) = $-0.0002X^2 + 0.1267X + 154.7277$, $R^2 = 0.8670$, $p < 0.001$). Significantly ($p < 0.05$) greater weight gain and SGR were observed in sea cucumber fed the NTS500 diet than those of sea cucumber fed all other diets (Con, LTS25, LTS50, LTS100, LTS250, LTS500, NTS25, NTS50, NTS100 and PTS100 diets), but not significantly ($p > 0.05$) different from those of sea cucumber fed the NTS250 diet. There was no significant ($p > 0.05$) difference in growth (weight gain and SGR) of sea cucumber fed the NTS100 and PTS100 diets.

3.2 | The proximate composition of the whole body of sea cucumber

The whole-body moisture content ranged from 91.3 to 91.7% and crude protein content ranged from 3.3 to 3.5% (Table 3). The whole-body crude lipid content was 0.3% for all diets and ash content ranged from 3.4 to 3.6%. These values were not significantly ($p > 0.05$) affected by either sulphur type or sulphur concentration.

3.3 | Cumulative mortality of sea cucumber subjected to air exposure stress

The cumulative mortality of sea cucumber was significantly affected by both dietary sulphur type ($p < 0.03$) and concentration ($p < 0.0001$) at 24 h after the 30-h air exposure stress (Table 4). However, dietary sulphur type and concentration did not affect mortality of sea cucumber at 48 h after the 30-h air exposure. At 72 and 96 h, dietary sulphur type significantly ($p < 0.02$) affected mortality of sea cucumber, but not for dietary sulphur concentration.

No mortality of sea cucumber was observed during the 30-h air exposure stressor, but significantly ($p < 0.05$) higher mortality was observed in sea cucumber fed the commercial feed compared with that of sea cucumber fed all other diets at 6 h after the 30-h air exposure throughout the 4-day post observation (Figure 1). Sea cucumber fed the Con diet showed significantly ($p < 0.05$) higher cumulative mortality than that of sea cucumber fed the NTS50, NTS100, NTS250, NTS500 and LTS500 diets, but not significantly ($p > 0.05$) different from that of sea cucumber fed the LTS25, LTS50, LTS100, LTS250, NTS25 and PTS100 diets at 78 h after the 30-h air exposure until the end of the 4-day post observation. Significantly ($p < 0.05$) lower cumulative mortality of sea cucumber fed the NTS250 and NTS500 diets compared with that of sea cucumber fed the Con and LTS25 diets was observed at the end of 4-day post observation. The lowest mortality was obtained in sea cucumber fed the NTS500 diet at the end of 4-day post observation after the 30-h air exposure.

3.4 | Cumulative mortality of sea cucumber subjected to low salinity exposure stress

The cumulative mortality of sea cucumber was significantly affected by both dietary sulphur type ($p < 0.001$, $p < 0.0001$, $p < 0.0001$ and $p < 0.0001$) and concentration ($p < 0.02$, $p < 0.001$, $p < 0.0001$ and $p < 0.006$) at 24, 48, 72 and 96 h, respectively, after the 12-h low salinity exposure stress (Table 4).

Sea cucumbers fed all diets were alive during the 12-h low salinity stress (Figure 2). Significantly ($p < 0.05$) higher cumulative mortality was observed in sea cucumber fed the Con, LTS25, LTS100 and commercial diet compared with that of sea cucumber fed the LTS500, NTS25, NTS500 and PTS100 diets, but not significantly ($p > 0.05$) different from that of sea cucumber fed the

TABLE 3 Chemical composition (%) of the whole body of juvenile sea cucumber fed the experimental diets for 8 weeks.

Experimental diets	Moisture	Crude protein	Crude lipid	Ash
Con	91.5 ± 0.12	3.4 ± 0.07	0.3 ± 0.03	3.5 ± 0.07
LTS25	91.7 ± 0.03	3.4 ± 0.03	0.3 ± 0.02	3.5 ± 0.07
LTS50	91.4 ± 0.17	3.4 ± 0.02	0.3 ± 0.02	3.6 ± 0.10
LTS100	91.3 ± 0.13	3.4 ± 0.09	0.3 ± 0.01	3.6 ± 0.17
LTS250	91.3 ± 0.04	3.3 ± 0.08	0.3 ± 0.03	3.5 ± 0.05
LTS500	91.4 ± 0.08	3.4 ± 0.03	0.3 ± 0.01	3.5 ± 0.05
NTS25	91.3 ± 0.08	3.4 ± 0.02	0.3 ± 0.02	3.6 ± 0.09
NTS50	91.4 ± 0.07	3.4 ± 0.07	0.3 ± 0.04	3.5 ± 0.06
NTS100	91.3 ± 0.22	3.5 ± 0.06	0.3 ± 0.02	3.5 ± 0.10
NTS250	91.5 ± 0.24[a]	3.4 ± 0.01[a]	0.3 ± 0.00[a]	3.4 ± 0.06[a]
NTS500	91.3 ± 0.12[a]	3.3 ± 0.13[a]	0.3 ± 0.01[a]	3.5 ± 0.08[a]
PTS100	91.4 ± 0.05[a]	3.4 ± 0.17[a]	0.3 ± 0.04[a]	3.4 ± 0.08[a]
Two-way ANOVA				
Type	$p > .5$	$p > .8$	$p > .8$	$p > .9$
Concentration	$p > .6$	$p > .4$	$p > .3$	$p > .8$
Interaction	$p > .3$	$p > .9$	$p > .4$	$p > .9$

Values (means of triplicate ±SE) in the same column sharing the same superscript letters are not significantly different ($p > 0.05$).

TABLE 4 Results of two-way analysis of variance to monitor dietary effect of sulphur type and concentration on cumulative mortality (%) of sea cucumber subjected to the 30-h air and 12-h low salinity stressors during the 96-h post observation periods.

Source of variation	Air exposure stressor				Low salinity stressor			
Elapsed hours after returning sea cucumber to seawater[a]	24	48	72	96	24	48	72	96
Type	$p < 0.03$	$p > 0.06$	$p < 0.02$	$p < 0.02$	$p < 0.001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$
Concentration	$p < 0.0001$	$p > 0.06$	$p > 0.06$	$p > 0.1$	$p < 0.02$	$p < 0.001$	$p < 0.0001$	$p < 0.006$
Interaction	$p > 0.7$	$p > 0.5$	$p > 0.9$	$p > 10.0$	$p > 0.2$	$p > 0.7$	$p > 0.3$	$p > 0.9$

[a]1, 2, 3 and 4-day post observation periods after the 30-h air exposure stress and 12-h low salinity exposure stress indicate 54, 78, 102 and 126 h (Figure 1), and 36, 60, 84 and 108 h (Figure 2), respectively.

LTS50, LTS250, NTS50, NTS100 and NTS250 diets at 18 h after the 12-h low salinity stress. The highest cumulative mortality was observed in sea cucumber fed the commercial diet at 33 h and lowest for sea cucumber fed the NTS500 diet at 24 h after the 12-h low salinity stress until the end of 96-h post observation. Sea cucumber fed the Con diet showed significantly ($p < 0.05$) higher cumulative mortality than that of sea cucumber fed all NTS, PTS100 and LTS500 diets, but not significantly ($p > 0.05$) different from that of sea cucumber fed the LTS25, LTS50, LTS100 and LTS250 diets at 60 h after the 12-h low salinity stress until the end of the 96-h post observation.

4 | DISCUSSION

Body metabolic processes such as growth in weight and length and immune activity are regulated by availability of the essential amino acids including sulphur-containing amino acids (methionine and cysteine) (Pazla et al., 2018). Therefore, sulphur-rich ingredient seems to enhance growth and immune response of animal in feed, and to the best of our knowledge, it is the first report on effect of sulphur type and concentration on survival and growth performance of invertebrate aquatic animal, especially sea cucumber. However, red drum (*Sciaenops ocellatus*) were able to use the different sulphur amino acid compounds, such as DL-methionine, crystalline methionine hydroxyl analog, N-acetyl-DL-methionine or Alimet™ (a liquid methionine hydroxyl analog) equally as effective as L-methionine to satisfy their sulphur amino acid requirement (Goff & Gatlin, 2004). They also reported that cystine was able to spare approximately 50% of the dietary methionine requirement of the fish.

Greater weight gain and SGR of sea cucumber fed all NTS diets and PTS100 diet compared with those of sea cucumber fed all LTS diets, except for the LTS500 diet in this study indicated that administration of nano-powder and powder type sulphur seemed to be more effective than liquid type sulphur for sea cucumber. This could

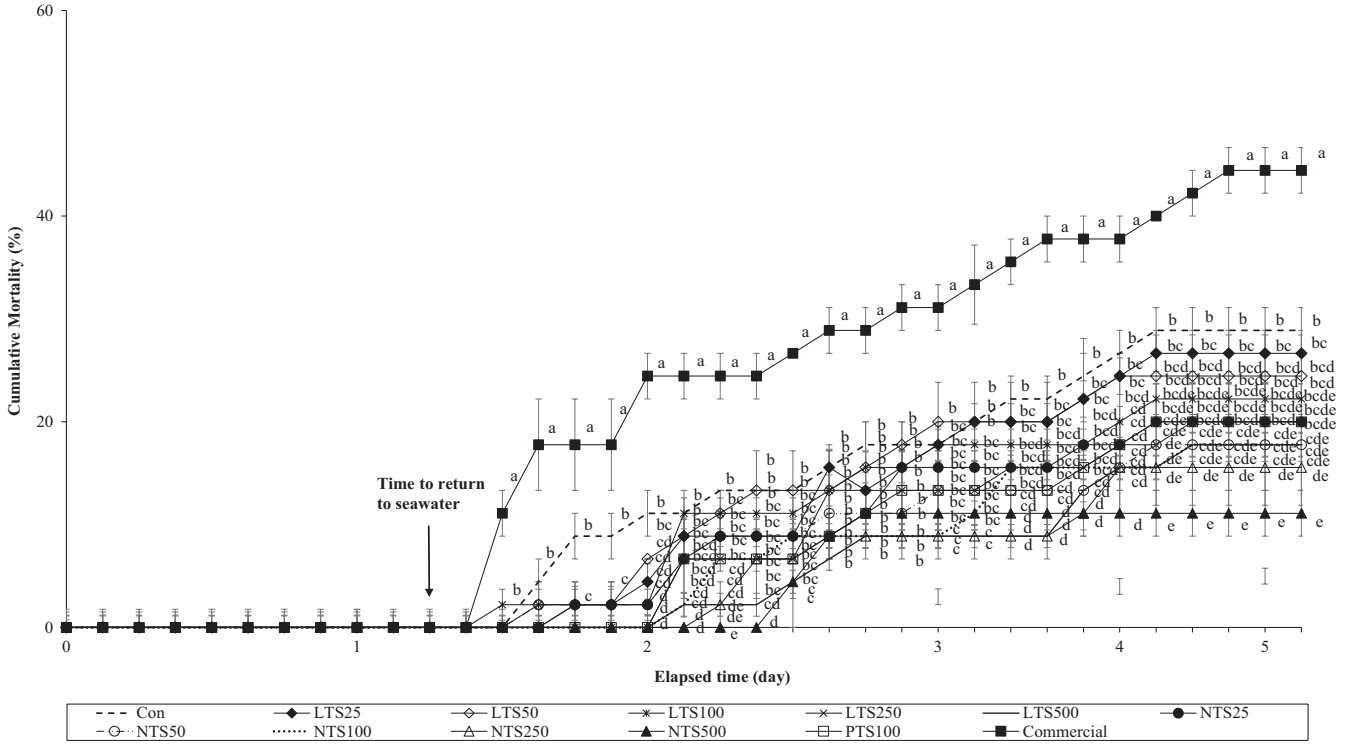

FIGURE 1 Cumulative mortality (%) of sea cucumber during the 4-day post observation periods after the 30-h air exposure stress at the end of the 8-week feeding trial. Values (means of triplicate ±SE) in the same elapsed time sharing the same superscript letters are not significantly different ($p > .05$).

be resulted from easy dilution (leaching out) of sulphur to surroundings (seawater) in the latter. However, no significant difference in weight gain and SGR of sea cucumber fed the NTS100 and PTS100 diets indicated that powder type sulphur did not affect performance of sea cucumber in this experimental condition. Dietary increased amount of nano-sulphur content in diets tended to improve weight gain (SGR) of sea cucumber and the maximum weight gain of sea cucumber was estimated to be 316.8 ppm of NTS content based on the 2nd order regression analysis of weight gain and NTS in diets in this study. Similarly, Lim et al. (2018) showed that oral supplementation of organic sulphur affected the performance and egg quality of laying hens and stimulated immune response when 31-week-old hens were fed with one of the diets containing four levels (0, 0.1, 0.2 and 0.4%) of organic sulphur for 24 weeks. Li et al. (2009) also reported that growth performance indices (survival, weight gain and feed efficiency) of juvenile hybrid striped bass (*Morone chrysops* ×M.*saxatilis*) tended to improve by dietary increased concentration of DL-methionine or DL-2 hydroxy-4-methylthiobutanoic acid in the 6-week feeding trial, and assumed that reserve antioxidants, such as ascorbic acid in fish tissue may reduce when methionine (sulphur amino acid) is deficient in feed, which may result in oxidative stress, poor growth performance and high mortality. Therefore, administration of sulphur, especially NTS and PTS rather than LTS can impose positive effect on growth performance of sea cucumber.

The chemical composition of the whole body of sea cucumber was not affected by either dietary sulphur type or sulphur concentration in this study. Unlike this study, however, dietary sufficiency of sulphur amino acid compounds changed the whole-body moisture and protein contents of hybrid striped bass (Li et al., 2009). Dietary inclusion of different sulphur amino acids compounds also changed the whole-body composition (dry matter, crude protein and lipid content) and muscle protein content of red drum (Goff & Gatlin, 2004). Crude protein and lipid content of the whole body of juvenile rock bream (*Oplegnathus fasciatus*) gradually increased and decreased, respectively, with the addition of taurine in the presence of sulphur amino acids in diets (Ferreira et al., 2015).

Nowadays, the invertebrates cultured in different farms should monitor regularly and supply a nutrition-balanced diet to reduce impact of inevitable stressors during culturing, such as air exposure stressor resulted from size grading, cleaning and transporting and low salinity exposure stressor during the rainy season. The cumulative mortality of sea cucumber was affected by both sulphur type and concentration at 24 h after 30-h air exposure, but by the sulphur type only at 72 and 96 h during the 96-h post observation (Table 4). The cumulative mortality of sea cucumber was affected by both sulphur type and concentration after the 12-h low salinity exposure at 24, 48, 72 and 96 h throughout the 96-h post observation. Improved survival of sea cucumber treated by sulphur and then subjected to either 36-h air exposure or 12-h low salinity exposure indicated that both sulphur type, especially NTS and/or sulphur concentration have desirable effect on lowering mortality of sea cucumber subjected to abiotic stressors.

Lowest mortality was obtained in sea cucumber fed the NTS500 diet at the end of 96-h observation after both the 30-h air and 12-h low salinity exposures, but highest for sea cucumber fed the

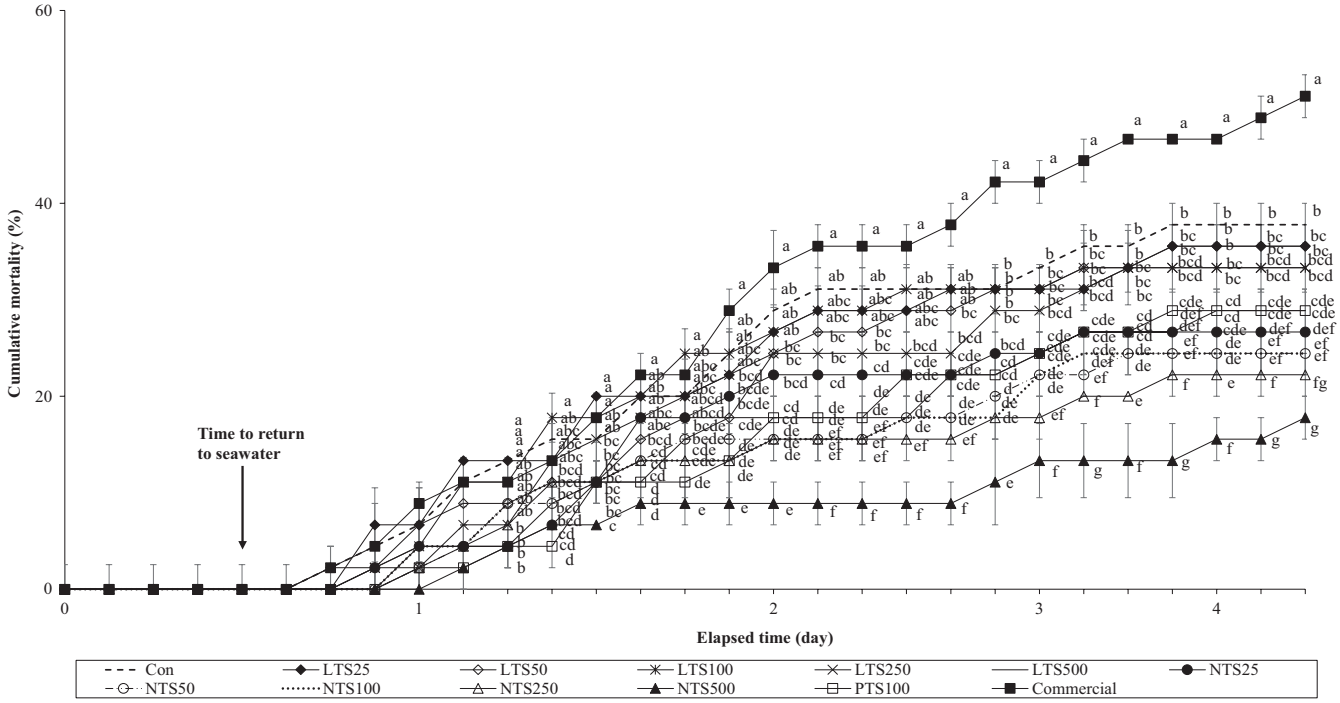

FIGURE 2 Cumulative mortality (%) of sea cucumber during the 4-day post observation periods after the 12-h low salinity stress at the end of the 8-week feeding trial. Values (means of triplicate ±SE) in the same elapsed time sharing the same superscript letters are not significantly different ($p > 0.05$).

commercial diet. Relatively high cumulative mortality of sea cucumber fed the Con diet at the end of the 96-h post observation after both the 30-h air exposure and 12-h low salinity exposures indicated that sea cucumber fed the diets treated by sulphur regardless of sulphur type and concentration seemed to more capable to resist both air and low salinity exposure stressors. Similarly, aquatic animals, such as fish and molluscs fed nutritionally balanced diets, have higher capacity to survive against abiotic stressors compared with those fed nutritionally unbalanced diets (Ansary, Baek, et al., 2019; Ansary, Jeong, et al., 2019; Duan et al., 2016; Lee et al., 2016; Liu et al., 2015).

In addition, lower mortality of sea cucumber fed all NTS diets, except for the NTS25 diet compared with that of sea cucumber fed the Con diet at 78 h after the 30-h air exposure until the end of the 96-h post observation in this study specified that oral administration of NTS may improve immunity and the physiological health of sea cucumber against air exposure stressor. Hou et al. (2019) described that the immunological parameters of sea cucumber were affected by desiccation stress. Air exposure affects the survival capacity of aquatic animals (Xu et al., 2018) although their tolerance capacities largely depend on the size, health status and species of the aquatic animals (Cardinaud et al., 2014; Wells & Baldwin, 1995). Sea cucumber inhabiting in the shallow seawater undergoes daily and seasonal salinity fluctuations (Wang et al., 2008), which affects the osmotic pressure, ion homeostasis and Na^+/K^+-ATPase activity of sea cucumber (Geng et al. 2016). Sulphur supplementation increasing the beneficial microbial population may have positive effect on growth and immunity of animal (Pazla et al., 2018). Furthermore, Espe et al. (2016) reported that Atlantic salmon (*Salmo salar*) fed the low-methionine diet faced higher metabolic stress compared with fish fed the proper-methionine diet. The main function of sulphur is to give support to the formation of amino acids containing sulphur and also some vitamins and co-enzymes (Kandylis, 1984), which might be involved in improvement in non-specific immunity against different abiotic stressors. Supplementation effect of LTS on growth and stress resistance of sea cucumber was poorer than dietary inclusion of NTS. Dietary inclusion of NTS should be carefully considered rather than supplementation of LTS in terms of improved growth and stress resistance of sea cucumber as well as less sulphur pollution source in effluent discharged from sea cucumber farm. Feasibility of supplementation of sulphur in commercial scale sea cucumber farm should be tested before prior to practical application.

5 | CONCLUSION

Both sulphur type and concentration in diets have desirable effect on growth performance (SGR) of sea cucumber. Regardless of sulphur concentration, all NTS and PTS 100 diets achieved superior weight gain and SGR of sea cucumber to all LTS diets, except for the LTS500 diet, and Con diet. Weight gain and SGR of sea cucumber tended to improve with the increased amount of NTS in diets. Sea cucumber fed the Con diet achieved higher cumulative mortality than that of sea cucumber fed the NTS50, NTS100, NTS250, NTS500 and LTS500 diets at 78 h after the 30-h air exposure. Higher cumulative mortality was observed in

sea cucumber fed the Con, LTS25, LTS100 and commercial diets compared with sea cucumber fed the LTS500, NTS25, NTS500 and PTS100 diets at 18 h after the 12-h low salinity exposure. Supplementation of sulphur, especially NTS has desirable effect on growth and stress reducer of sea cucumber subjected to air and low salinity exposures.

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CONFLICT OF INTEREST

The authors have no conflict of interest to declare.

AUTHOR CONTRIBUTIONS

JK and AY carried out the feeding trial and the chemical analysis of data. SHC designed the study and prepared the manuscript. CHK funded the research.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author.

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